

Reza Ahmadi

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Research Interests

Software Engineering, Program Analysis, Cybersecurity, Machine Learning, Deep Learning, Computer Vision, NLP

Education

University Of Verona

Master's degree in Computer Engineering

Specialization: Robotics and Smart Industry. Average: 100/110.

Thesis: "Automated API Specification Inference from Network Traffic of Android Applications"

Verona, Italy

2023–2026

Shahab Danesh University

Bachelor's degree in Software Engineering

Selected as 2nd Top Student in the Software Engineering department (2017). Average: 16.78 / 20

Qom, Iran

2016–2018

Kashan Boys Technical College

Associate degree in Software Engineering

Average: 14.43 / 20.

Kashan, Iran

2013–2015

Research Experience

Research Fellow (AI & Software Development)

University of Verona

Verona, Italy

Apr 2026–Present

Selected for a competitive research fellowship on the "FINAL LAP" project, externally funded by **Fondazione Cariverona**, under the supervision of Prof. Marco Cristani. Member of **INTELLIGO Labs** — AI for Materials (AI4M) research group.

- Designing and implementing AI-driven solutions spanning Physics-Informed Neural Networks (PINNs), Large Language Models (LLMs), and Computer Vision for intelligent material design.
- Developing full-stack software infrastructure to support research workflows, model training pipelines, and experimental evaluation.

R&D Software Engineer (Applied AI & Mobile)

Qualyco (University of Verona Spinoff)

Verona, Italy

Oct 2025–Mar 2026

- Engineered an applied computer vision solution for the **Benelli** manufacturing facility to optimize industrial warehouse operations.
- Designed and integrated a robust **Optical Character Recognition (OCR)** pipeline on Android edge devices to automate the tracking of high-precision components in real-time.

Research Fellow (Cybersecurity & IoT Systems)

University of Verona

Verona, Italy

Dec 2024–Jun 2025

Selected for a competitive, fully-funded research fellowship under the **iNEST** project, backed by the **European Union (NextGenerationEU)**.

- Conducted research on automated API specification inference from network traffic of Android applications (AndroidMITM) under Prof. Mariano Ceccato.
- Developed an automated massive testing framework on Docker and Podman with Python APIs and mitmdump.
- Automated testing processes for capturing and analyzing traffic via custom certificate integration on Android emulators.
- Configured Android SDK command-line tools for seamless network traffic interception and security audits.

Publications / Working Papers

2026: *Working Paper: Dynamic Inference of REST APIs from Android Network Traffic (Under Review)*

Reza Ahmadi, Sofia Mari, Michele Pasqua, Mariano Ceccato

Experience

Android Developer

Reply

Milan, Italy

Oct 2025–Jan 2026

- Contributed to the development and maintenance of UniCredit and Banco BPM banking applications.

Android Developer

Bale Messenger

Tehran, Iran

2020–2024

Bale is a banking messenger app serving over 15 million active users monthly.

- Revamped outdated modules and refactored 10,000+ lines of legacy code.
- Enhanced system memory performance by 40% and achieved 99% crash-free performance.
- Developed more than 10 new semi-major features and led performance optimization initiatives.

Android Developer

Basalam

Qom, Iran

2018–2020

Basalam is a marketplace with over 10 million users.

- Developed the application from scratch in 2 months, reaching 1 million users within 2 years.
- Awarded "Best Social Marketplace Application in Iran" (IWMF 2019).

Android Developer

MTN Irancell

Tehran, Iran

2017–2018

- Restructured codebase for maintainability and decreased technical debt by 20%.

Awards & Certificates

- The best social marketplace application of Iran (IWMF 2019) - Basalam
- 3rd Place in ACM Programming Competition, Shahab Danesh University (2017)
- IELTS (B2 Proficiency), IDP Education Ltd (2023)
- Android Development Professional Certificate, i3center (2016)

Skills

AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision

Cybersecurity & Networking: Mobile Security Auditing, MITM Analysis, Network Traffic Analysis, mitm-proxy/mitmdump, API Security, SSL/TLS Interception

Mobile Development: Android Native Development, Android SDK, Jetpack Compose

Software Architecture & Principles: OOP, SOLID, Clean Architecture, Reactive Programming, Data Structures & Algorithms, Performance Optimization

Core CS: Bash/Shell Scripting, Databases (SQL, SQLite, MySQL, NoSQL)

Programming Languages: Java, Kotlin, Python, C++, JavaScript

DevOps & Tools: Linux/Unix Environment, Docker, Podman, Git, CI/CD Pipelines, Gradle, LaTeX

Testing: Automated REST API Testing, Unit/Integration Testing, UI Testing

Languages

English: B2 (Professional Working Proficiency - Master's degree conducted in English)

Italian: A2 (Elementary)

Persian: Native